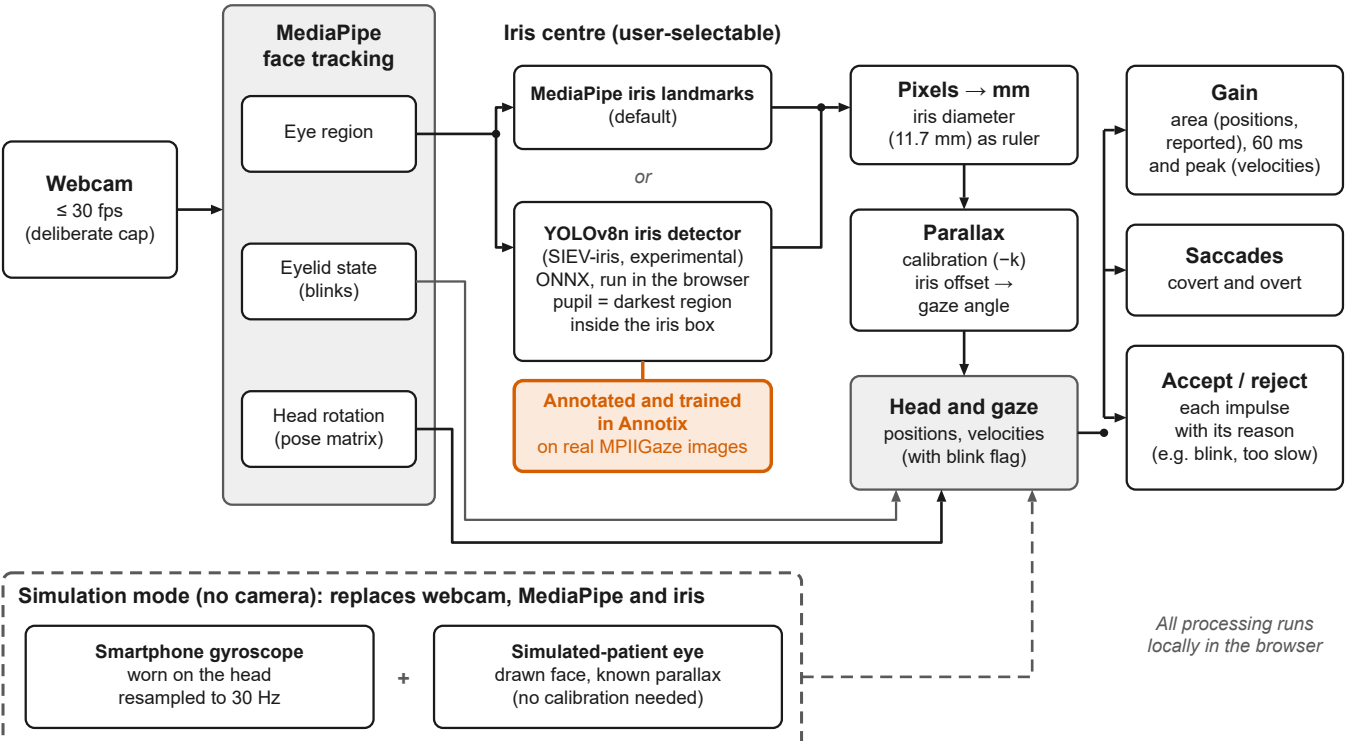

Webcam
≤ 30 fps
(deliberate cap)
MediaPipe face tracking
Eye region
Eyelid state (blinks)
Head rotation (pose matrix)
Iris centre (user-selectable)
MediaPipe iris landmarks (default)
or
YOLOv8n iris detector
(SIEV-iris, experimental)
ONNX, run in the browser
pupil = darkest region inside the iris box
Annotated and trained in Annotix
on real MPIIGaze images
Pixels → mm
iris diameter (11.7 mm) as ruler
Parallax
calibration (−k)
iris offset → gaze angle
Head and gaze
positions, velocities (with blink flag)
Gain
area (positions, reported), 60 ms and peak (velocities)
Saccades
covert and overt
Accept / reject
each impulse with its reason (e.g. blink, too slow)
Simulation mode (no camera): replaces webcam, MediaPipe and iris
Smartphone gyroscope
worn on the head
resampled to 30 Hz
+
Simulated-patient eye
drawn face, known parallax
(no calibration needed)
All processing runs locally in the browser